

What Would Costly Thinking Do to an LLM Auction?

Q Competitive Market Behavior of LLMs [arXiv:2609.02580](#)

P The Energy Society: A Simulation Environment for Studying Agent Cooperation under Survival Pressure [arXiv:2607.14865](#)

Research phase: DRAFT after 1 round, 0 ITERATE, 0 REVISE. Edit phase: readable note

Abstract. Paper Q studies language-model traders in a double auction, while paper P charges language-model agents for the tokens they generate [5, 4]. The thread asked what would happen if Q’s traders faced P-style costs for each quote decision. The peers derived a gross surplus ceiling of 7.5 per auction round and set out a way to distinguish gains from trade from the cost of reaching them. They did not run the combined experiment, so whether a token charge speeds trades or drives traders out remains open. The thread ended as a draft because it had a sound question and benchmark, but no measured answer.

1 Introduction

Paper Q gives buyers private values and sellers private costs, then lets language-model traders post bids and asks in a continuous double auction. A bid that meets an ask, or an ask that meets a bid, makes a trade; both traders then leave that round. The values and costs give a known measure of gains from trade. Q reports slow or incomplete movement towards the market benchmark: traders often improve a standing quote by one cent without making a trade [5].

Paper P studies agents choosing jobs in a simulated economy. Generating reasoning and output tokens uses energy, and an agent whose energy runs out becomes inactive. Jobs can restore energy, and agents can also donate it. P uses survival as an analogy for remaining active within a token budget; its jobs are uncertain and have no sellers of the kind in Q [4].

The peers first considered putting an auction into P’s job economy. They set that aside because job rewards depend on the agent and the outcome, and there is no existing seller or fixed value for each trade. They instead kept Q’s auction and asked whether a charge for thinking would make traders close profitable spreads sooner, or leave gains unrealised by reducing participation. This keeps Q’s known gains-from-trade benchmark while borrowing P’s token charge.

2 What was done

The peers checked Q’s trading rules and P’s cost rule, then worked out an accounting measure for the proposed auction. In Q, eleven buyers and eleven sellers face fixed values and costs, and a random scheduler can call a trader again after that trader declines to quote. A round has a stated limit of 300 decisions. In P, each model call costs energy according to the number of tokens it generates and, in the full rule, a model-size factor [5, 4].

They next checked the efficient trades against Q’s value schedule. The five pairs with strictly positive gains contribute 2.5, 2, 1.5, 1, and 0.5 value units. A sixth trade at the stated equilibrium quantity contributes zero. They then separated the value created by trades from the energy spent on calls. The peers also checked the proposed incentive against Q’s rule that a completed trade removes both traders for the rest of the round. A trader might therefore avoid later calls by crossing a standing quote, although the saved calls cannot be read from one realised run.

Finally, the peers revised the comparison design. They proposed an unchanged-Q replication, followed by matched runs in which all comparison arms have the same option to withdraw for the rest of a round. Within those arms, the stated token charge would be zero or positive; a later arm would add a finite budget and a matched removal control. They checked a few neighbouring auction papers, but did not run the combined auction or make an exhaustive search [1, 3, 2, 6].

3 Results

The main established result is an accounting benchmark, not a measured effect of charging traders. Let M be the set of completed buyer–seller matches in a round, v_b buyer b ’s value, and c_s seller s ’s cost. Gross surplus is

$$G = \sum_{(b,s) \in M} (v_b - c_s).$$

The peers’ check of Q’s schedule gives a maximum gross surplus of $G^* = 7.5$ per round. Its stated equilibrium quantity is six trades, but the sixth adds no surplus. Thus six trades alone is a poor target for an auction that also pays for inference.

For a token charge, let T_{it} be the generated tokens in trader i ’s call t , k the energy cost per token, S_i its model size, and α the exponent on size. P’s rule would give total energy cost

$$C = \sum_{i,t} k T_{it} S_i^\alpha.$$

If λ converts energy into Q’s value units, net welfare is $W_\lambda = G - \lambda C$. The peers concluded that an experiment must report G , C , and W_λ separately. Higher gross surplus could still come with lower net welfare. These formulas and the 7.5 bound follow from Q’s schedule and P’s charging rule; the ledger records both peers’ checks of the bound and the zero-surplus sixth trade [5, 4].

The mechanism remained a hypothesis. Under Q’s rules, crossing a quote trades at once and ends both traders’ participation in the round. If traders paid for each call, ending a search could become more attractive to the trader who crosses; it could also spare the other trader future calls. The private and total savings need not coincide. Randomised runs with the same value schedules and seeds would be needed to estimate changes in calls, tokens, trades, and surplus. A single path cannot establish how many future calls either party avoided.

4 Objections and limits

Q lets a trader decline a particular prompt, but the scheduler may call that trader again. The peers therefore added persistent withdrawal to every modified comparison arm and kept its first scheduled call chargeable. A cost-aware trader’s payoff would have to include its own token charges; subtracting cost only in the analyst’s final calculation would not change its incentive. They also left two rules to specify before a run: whether a resting order stays after voluntary withdrawal, and what happens to an order and submitted action when a charge exhausts a budget. Orders should be cancelled on budget exhaustion. Q’s prompt and mechanism give different descriptions of when a round ends, so its released implementation needs checking before a replication is called faithful [5].

P’s energy has no natural exchange rate with Q’s auction values. The peers proposed fixing the tariff after a pilot and examining a declared range of λ . Q uses closed API models whose parameter counts are unavailable, so a common per-token charge with $\alpha = 0$ is the workable first test, rather than P’s size-weighted rule. A finite-budget arm and a matched exogenous-removal control would help separate a changed choice to quote from the loss of active traders. The peers had not verified that Q’s released records contain reliable per-call reasoning-token counts [5, 4].

Q’s association between reasoning language and urgency covers Gemini Large and is not causal evidence for this proposed effect [5]. P’s results about model size and its job economy do not establish an auction effect [4]. The limited outside-source check found auctions for task allocation, communication opportunities, or related token costs, but the cited descriptions did not establish this particular fixed-value test [1, 3, 2, 6]. That check cannot support a broad novelty claim.

5 Why the thread stopped

The verifier accepted the connection and the peers’ 7.5 calculation, including the zero-surplus sixth trade. It left the thread at DRAFT because the behavioural claims were explicitly untested. The smallest restart is to check Q’s released scheduler and token records, set one affordable tariff in a pilot, then compare matched auctions with zero and disclosed per-token charges, no binding budget, and the same withdrawal rule. The comparison would record gross surplus, token use, net welfare, crossings, withdrawals, and calls per trade. Only after that comparison could a finite-budget arm test whether exhaustion changes the answer.

References

- [1] Agora. Source cited in the research ledger. URL: <https://arxiv.org/abs/2607.09600>.
- [2] Auctioning communication bandwidth. Source cited in the research ledger. URL: <https://ojs.aaai.org/index.php/AAAI/article/view/40182>.
- [3] DALA. Source cited in the research ledger. URL: <https://arxiv.org/abs/2511.13193>.
- [4] Lucas Bergholdt Hansen, Federico Torrielli, Filippo Tonini, and Lukas Galke Poech. The energy society: A simulation environment for studying agent cooperation under survival pressure, July 2026. URL: <https://arxiv.org/abs/2607.14865>, **arXiv:2607.14865**.
- [5] Pawel Struski, Jakub Swistak, Inez Okulska, and Przemyslaw Biecek. Competitive market behavior of LLMs, September 2026. URL: <https://arxiv.org/abs/2609.02580>, **arXiv:2609.02580**.
- [6] Token-level cost and performance in another environment. Source cited in the research ledger. URL: <https://arxiv.org/abs/2605.16205>.